

independence

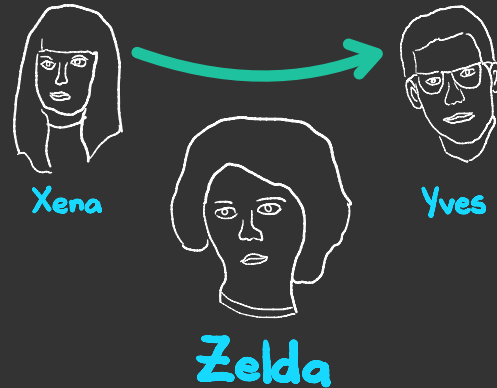
CSCI
270

the value of information is **conditional**

without Zelda's bloodtype,
Xena's bloodtype provides **no**
information about Yves'

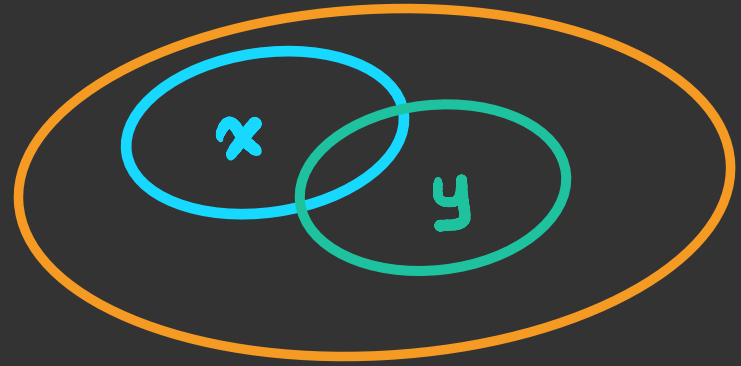


given Zelda's bloodtype,
Xena's bloodtype **could** provide
information about Yves'



let x be the event that
Xena has type O blood

let y be the event that
Yves has type AB blood



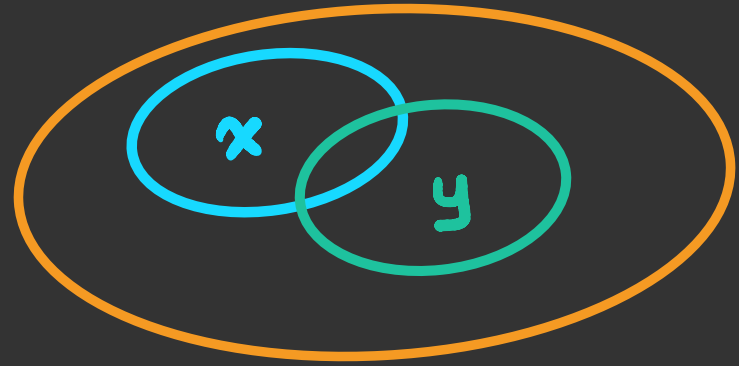
translate into probability notation:
"learning that Xena has type O blood
provides no information about
whether Yves has type AB blood"



???

let x be the event that
Xena has type O blood

let y be the event that
Yves has type AB blood



translate into probability notation:
"learning that Xena has type O blood
provides no information about
whether Yves has type AB blood"



$$P(y | x) = P(y)$$

events x and y are independent iff

$$P(y | x) = P(y)$$

independence of two events

$$\Rightarrow \frac{P(x \cap y)}{P(x)} = P(y) \Rightarrow$$

events x and y are independent iff

$$P(x \cap y) = P(x) P(y)$$

$$\Rightarrow \frac{P(x \cap y)}{P(y)} = P(x) \Rightarrow$$

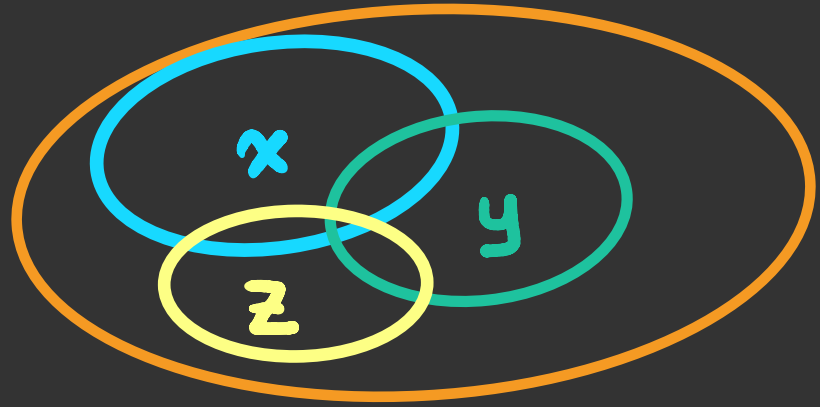
events x and y are independent iff

$$P(x | y) = P(x)$$

let x be the event that
Xena has type O blood

let y be the event that
Yves has type AB blood

let z be the event that
Zelda has type A blood



translate into probability notation:

"given we know **Zelda has type A blood**,
learning that **Xena has type O blood**
provides no information about
whether **Yves has type AB blood**"

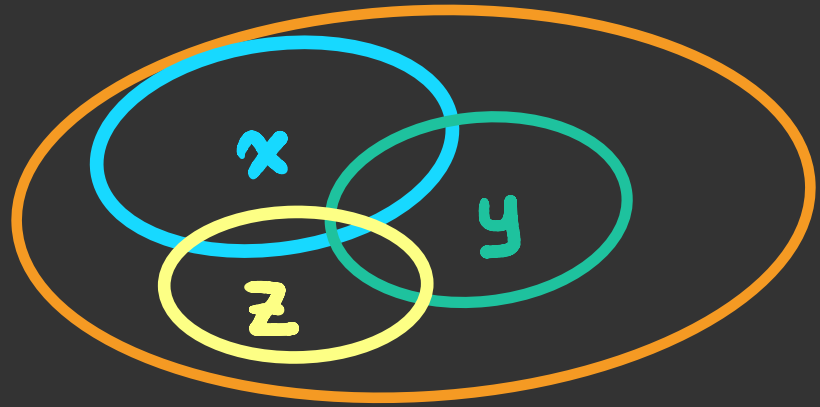


???

let x be the event that
Xena has type O blood

let y be the event that
Yves has type AB blood

let z be the event that
Zelda has type A blood



translate into probability notation:

"given we know **Zelda has type A blood**,
learning that **Xena has type O blood**
provides no information about
whether **Yves has type AB blood**"



$$P(y | x, z) = P(y | z)$$

conditional independence of events

events x and y are **conditionally independent** given event z iff

$$P(y | x, z) = P(y | z)$$

events x and y are **conditionally independent** given event z iff

$$P(x | y, z) = P(x | z)$$

events x and y are **conditionally independent** given event z iff

$$P(x \cap y | z) = P(x | z)P(y | z)$$



Xena



Yves



Zelda

let X be Xena's
blood type

let Y be Yves'
blood type

let Z be Zelda's
blood type



let X be Xena's
bloodtype



let Y be Yves'
bloodtype



let Z be Zelda's
bloodtype

translate from
english into
probability notation:

"Xena's bloodtype provides no
information about Yves' bloodtype."

???



let X be Xena's
bloodtype



let Y be Yves'
bloodtype



let Z be Zelda's
bloodtype

translate from
english into
probability notation:

"Xena's bloodtype provides no
information about Yves' bloodtype."

$$P(y | x) = P(y) \text{ for all } x \in \text{supp}(X) \text{ and } y \in \text{supp}(Y)$$

marginal independence

Y is marginally independent of X if

$$P(y | x) = P(y) \text{ for all } x \in \text{supp}(X) \text{ and } y \in \text{supp}(Y)$$

denoted

$$Y \perp\!\!\!\perp X$$



Y is marginally independent of X if

$$P(y | x) = P(y) \text{ for all } x \in \text{supp}(X) \text{ and } y \in \text{supp}(Y)$$



let X be Xena's
bloodtype



let Y be Yves'
bloodtype



let Z be Zelda's
bloodtype

translate from
english into
probability notation:

"Xena's bloodtype provides no
information about Yves' bloodtype,
given that we know Zelda's bloodtype."

???



let X be Xena's
bloodtype



let Y be Yves'
bloodtype



let Z be Zelda's
bloodtype

translate from
english into
probability notation:

"Xena's bloodtype provides no
information about Yves' bloodtype,
given that we know Zelda's bloodtype."

$$P(y | x, z) = P(y | z)$$

for all $x \in \text{supp}(X)$, $y \in \text{supp}(Y)$, and $z \in \text{supp}(Z)$

conditional independence

Y is conditionally independent of X given Z if

$$P(y | x, z) = P(y | z)$$

for all $x \in \text{supp}(X)$, $y \in \text{supp}(Y)$, and $z \in \text{supp}(Z)$

denoted

$$Y \perp\!\!\!\perp X \mid Z$$



Y is conditionally independent of X given Z if

$$P(y \mid x, z) = P(y \mid z)$$

for all $x \in \text{supp}(X)$, $y \in \text{supp}(Y)$, and $z \in \text{supp}(Z)$

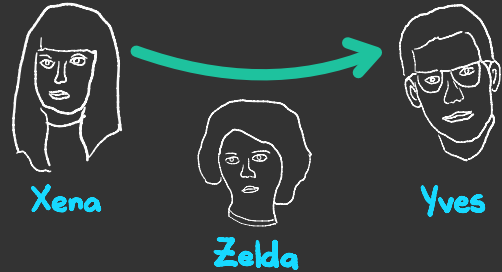
without Zelda's bloodtype,
Xena's bloodtype provides **no**
information about Yves'



so:

$Y \perp\!\!\!\perp X$

given Zelda's bloodtype,
Xena's bloodtype **could** provide
information about Yves'



so:

$Y \not\perp\!\!\!\perp X \mid Z$